BRAC University

Department of Mathematics and Natural Sciences

Lecture on

Real Analysis II (MAT321)

Continuity in Metric Space

Continuity, Uniform continuity, Homeomorphisms

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Conducted by

Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

partho.dhor@bracu.ac.bd | parthosutradhor@gmail.com

For updates subscribe on @ParthoSutraDhor

Continuity of Functions in Metric Spaces

Continuity at a Point in $\mathbb{R}$

Let $f : A \to \mathbb{R}$ be a function defined on a subset A of $\mathbb{R}$, and let $c \in A$. We say that f is **continuous at the point** a if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$\forall x \in A \text{ and } |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

Continuity at a Point in Metric Space

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$ be a function. We say that f is **continuous at the point** $a \in X$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$\forall x \in X \text{ and } d_X(x, a) < \delta \implies d_Y(f(x), f(a)) < \varepsilon.$$

Equivalent Definition using Open Spheres

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$ be a function. We say that f is **continuous at the point** $a \in X$ if for every open sphere $S_\varepsilon(f(a))$ in Y, there exists an open sphere $S_\delta(a)$ in X such that

$$f(S_\delta(a)) \subseteq S_\varepsilon(f(a)).$$

Equivalent Definition using Open Set

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$ be a function. We say that f is **continuous at the point** $a \in X$ if for every open set U in Y containing $f(a)$, there exists an open set V in X containing a such that

$$f(V) \subseteq U.$$

Continuous Functions

Continuity using Open Sets

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$ be a function. We say that f is continuous at all points of X i.e. **continuous** if for any open set U in Y, the preimage $f^{-1}(U)$ is an open set in X.

Continuity using Closed Sets

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$ be a function. We say that f is continuous at all points of X i.e. **continuous** if for any closed set C in Y, the preimage $f^{-1}(C)$ is a closed set in X.

Examples of Continuous Functions

Continuity in Discrete Metric Space

Let (X, d_X) be a discrete metric space, and let (Y, d_Y) be any metric space. Then every function $f : X \to Y$ is continuous at every point of X.

Continuity of a constant function

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$ be a constant function defined by $f(x) = c$ for all $x \in X$, where $c \in Y$ is a fixed point. Then f is continuous at every point of X.

Continuity using Sequences

Continuity at a Point using Sequences

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$ be a function. We say that f is **continuous at the point** $a \in X$ if and only if for every sequence (x_n) in X that converges to a, the sequence $(f(x_n))$ in Y converges to $f(a)$.

Uniform Continuity

Uniform Continuity

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$ be a function. We say that f is **uniformly continuous** if for every $\varepsilon > 0$, there exists a $\delta > 0$, that depends only on ε, such that

$$\forall x, y \in X \text{ and } d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon.$$

Cauchy sequence with uniform continuity

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$ be a uniformly continuous function. Then for every Cauchy sequence (x_n) in X, the sequence $(f(x_n))$ in Y is also a Cauchy sequence.

Not uniform continuity with Cauchy sequence

Give an example of a function $f : X \to Y$ that is continuous but not uniformly continuous, and a Cauchy sequence (x_n) in X such that the sequence $(f(x_n))$ is not a Cauchy sequence in Y.

Homeomorphisms

Homeomorphism

Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : X \to Y$ is called a **homeomorphism** if it is a bijection and both f and its inverse $f^{-1} : Y \to X$ are continuous.

Homeomorphic Spaces

Let (X, d_X) and (Y, d_Y) be metric spaces. We say that X and Y are **homeomorphic** if there exists a homeomorphism $f : X \to Y$. In this case, we write $X \cong Y$.

Thank You!

We'd love your questions and feedback.

Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

partho.dhor@bracu.ac.bd | parthosutradhor@gmail.com

@ParthoSutraDhor

(Lectures, walkthroughs, and course updates)

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References

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[2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.